

Dawson Verley

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Education

University of California, Berkeley	Berkeley, CA
M.S. in Energy and Resources	August 2024 — May 2026
• Emphasis: Machine Learning, Economics and Energy System Modeling • Relevant Coursework: Data Science for Energy, Energy System Modeling, Statistical Learning Theory, Probability Theory, Model Evaluation, Mathematical Economics, Climate Change Economics, Quantitative Policy Analysis	
Stanford University	Stanford, CA
B.A. in Political Science (Honors and Distinction, Phi Beta Kappa)	September 2018 — June 2022
• Emphasis: Data Science and Causal Inference • Relevant Coursework: Deep Learning, Statistical Learning, ML for Social Scientists, Data Science, Scientific Python • Research Affiliations: Golub Capital Social Impact Lab, King Center, Europe Center, Immigration Policy Lab	

Professional Experience

Graduate Student Instructor (Data Science)	August 2024 — May 2026
University of California, Berkeley	Berkeley, CA
• Introduced 500+ students to data science, machine learning and python programming • Developed course materials covering model selection, model evaluation, geospatial machine learning, and other topics	
Visiting Researcher	June 2025 — August 2025
Division of Energy Systems, KTH Royal Institute of Technology	Stockholm, Sweden
• Leveraged geospatial foundation models to estimate household fuel use trends across Sub-Saharan Africa • Designed and built an AI agent to accelerate systematic literature reviews in global public health	
Research Data Analyst	September 2023 — September 2024
Department of Political Science, Stanford University	Stanford, CA
• Built machine learning tools to harmonize large, heterogeneous survey datasets, dramatically accelerating research • Used xarray and dask to process terabyte-scale spatiotemporal data on cloud infrastructure	
Legislative Aide (Energy and Natural Resources)	January 2023 — January 2024
Alaska State House	Juneau, AK
• Translated complex energy policy topics into concise memos, briefings, and recommendations for legislators • Interfaced with advocacy organizations, industry groups, and government officials • Primary author of two bills passed into law	
Research Fellow (Computational Science)	August 2022 — January 2023
Stanford Law School	Stanford, CA
• Conducted large-scale computational analysis of California environmental policies using R and Python • Contributed to research on federal data privacy laws	

Selected Projects

Power Purchasing Agreement (PPA) Analytics Dashboard – PowerPurchasing.work	2026
• Built an ETL pipeline for an electricity market analytics dashboard, using DuckDB and SQL to query FERC EQR data • Designed and deployed a responsive web app for visualization and analysis of PPA price trends using JavaScript (Vite).	
Uncertainty Quantification for Energy Markets	2025
• Developed conformal prediction methods to quantify uncertainty in net load forecasts for wholesale energy markets • Benchmarked forecast intervals against standard engineering heuristics to improve decision-making	

Technical Skills and Proficiencies

- **Programming:** Python (pandas, polars, pytorch, numpy, xarray, dask), SQL, R (tidyverse), Java, JavaScript (React, Vite)
- **Data & Infrastructure:** Git, GitHub, Docker, CI/CD, ETL/ELT, AWS, GCP, Airflow, Databricks, BigQuery
- **Energy Markets:** SCUC, SCED, SCOPF, Capacity Expansion Modeling, Load Forecasting, Interconnection, PPAs